WENTWORTH LIU

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Data Scientist | Battery Health & Safety | Applied ML & Data Products

SUMMARY

Data Scientist in battery intelligence combining statistical inference, applied machine learning and engineering to support battery longevity and safety decisions. At Li Auto, partnered with BMS, cell-design, performance and after-sales teams on state-of-health forecasting, OTA strategy evaluation and explainable risk ranking. Built reusable feature infrastructure and production ML pipelines; previously delivered multi-source measurement tools at ByteDance. Brings computational materials science research and cross-functional software delivery experience.

EXPERIENCE

Li Auto *Jul 2024 – Present*
Data Scientist, Battery Intelligence Department

- **Behavioural Risk Scoring & MLOps.** Built an explainable relative-risk system to prioritise rare thermal-runaway investigations under extreme class imbalance. Captured 93% of evaluated cases within the top-ranked 2% of vehicles, narrowing investigation candidates from tens of thousands to thousands; the score ranks cases, not failure probabilities. Applied Mann–Whitney screening and correlation-cluster redundancy control to 1.5k+ vehicle-behaviour features, then developed decomposable, directional CDF-based scoring with platform-aware normalisation; productionised monthly Docker-packaged scoring with API delivery, validation, monitoring and case-driven refresh.
- **Causal Inference & Cohort-targeted Longevity Strategy.** Turned observational vehicle-behaviour data into cohort-targeted interventions by linking causal effect estimation, battery-degradation forecasting and OTA strategy evaluation. Used confounder control, propensity-score matching, ATT and dose-response estimation to isolate actionable drivers; recursively projected early-life battery state-of-health trajectories to identify warranty-risk cohorts. Partnered with BMS, cell-design, performance and after-sales teams to map feasible interventions and evaluate staged OTA rollout. Six-month matched-cohort and within-vehicle counterfactual evaluations supported a *projection* of tens of millions of RMB in avoided warranty costs for a single vehicle model, not realised savings.
- **End-to-End Offline Feature Store.** Designed 1.5k+ vehicle-cycle behaviour features and built an in-house, department-wide offline Feature Store, cutting feature-development time by approximately 85% and reducing duplicated computation and storage. Implemented a Git-managed YAML feature registry with metadata and lineage, schema-driven PySpark ETL with temporal alignment, validated ingestion, historical backfills, monitoring and model-ready export through internal APIs, making features reproducible across analysis, training and batch scoring.
- **Confidence-Aware NLP Work-Order Classification.** Built a confidence-aware system for 50k+ noisy after-sales work orders across 30+ categories, routing low-confidence cases to expert review. Automated over 93% of labelled samples, raising accuracy from 83% to 96% in three months (weighted F1 > 0.90) and replacing roughly 12 manual hours weekly. Co-designed the label taxonomy, evaluated expert terms using keyword-level F1, engineered domain-informed TF-IDF features and trained a Random Forest classifier with weekly retraining on verified labels.

ByteDance *May 2023 – Sep 2023*
Data Analytics

- **Fanqie IP Popularity Model for Promotion Prioritisation.** Built a daily model to inform which IPs warranted continued promotion. Integrated consumption, dissemination and search signals through automated ETL and data-quality monitoring; combined PCA-derived statistical weights with business-agreed AHP weights and cloud-deployed the dashboard. Reached approximately 60% internal adoption; a business retrospective recorded approximately 40% higher anchor-tracked app downloads across team-promoted videos in the month after marketing adjusted promotion priorities versus the preceding month—an observational comparison, not a measured causal effect of the model.
- **Qishui Music Audience Saturation & Acquisition Targeting.** Diagnosed falling anchor-linked new-user activations as audience saturation in video-led acquisition. Used funnel-stage analysis and cross-product audience matching to compare installed-user penetration with anchor-linked new-user activation rates by content vertical. Marketing shifted promotion from saturated to higher-potential verticals; no unverified uplift is claimed.

EDUCATION

Nanjing University *Sep 2021 – Jun 2024*
MSc Computational Materials Science. Built an open-source 3D Fermi-surface analysis tool; first author in *Physical Review B* (2024), co-first author in *Physical Review Materials* (2023); named inventor on a related patent granted in 2025.

Nanjing University *Sep 2017 – Jul 2021*
BSc Materials Science & Engineering. GPA 4.6/5.0; top 1%; National Scholarship; Outstanding Graduate; Jiangsu Province Outstanding Student; MCM/ICM 2020 Honorable Mention (team captain).

SKILLS

Programming: Python; SQL; PySpark; Git; Docker.
Modelling: Statistical Modelling; Causal Inference / PSM; Feature Engineering; Risk Ranking; NLP.
Delivery: MLOps; Feature Stores; Batch Pipelines; Data Validation; Model Monitoring.
Battery: Battery Longevity; State-of-Health Forecasting; Safety Risk Ranking; OTA Strategy Evaluation.
Languages: English (TOEFL 104); Mandarin/Cantonese (Native); Japanese (JLPT N1).